

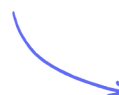
Exam Questions

3D Pythagoras & Trigonometry

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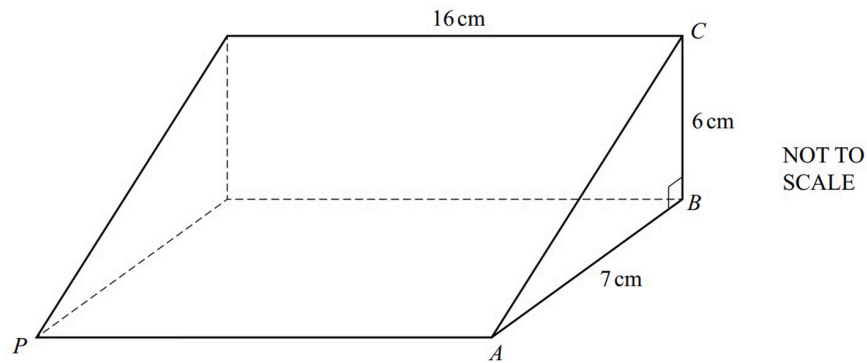
Medium (11 questions)	/38
Hard (11 questions)	/52
Very Hard (11 questions)	/53
Total Marks	/143

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Medium Questions

- 1 The diagram shows a prism of length 16 cm.



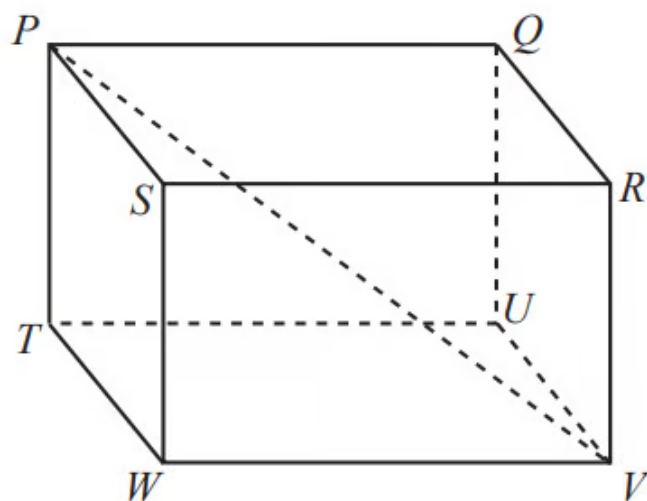
Triangle ABC is a cross-section of the prism with $AB = 7$ cm, $BC = 6$ cm and angle $ABC = 90^\circ$.

Calculate the length of PC

..... cm

(3 marks)

2



NOT TO
SCALE

The diagram shows a cuboid $PQRSTU VW$.

$$PV = 17.2 \text{ cm}$$

The angle between the line PV and the base $TUVW$ of the cuboid is 43° . Calculate PT .

$$PT = \dots\dots\dots \text{ cm}$$

(3 marks)

3 The diagram shows a cuboid $ABCDEFGH$.

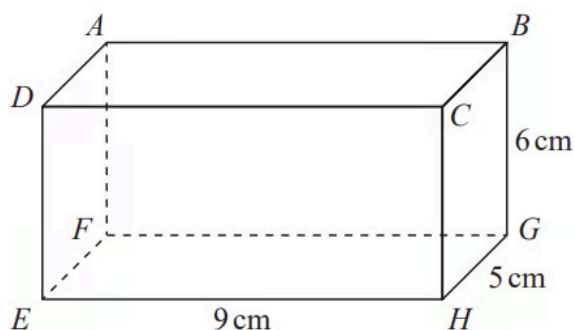


Diagram **NOT**
accurately drawn

$$EH = 9 \text{ cm}, \quad HG = 5 \text{ cm} \text{ and } \quad GB = 6 \text{ cm}.$$

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

(4 marks)

4 The diagram shows a triangular prism.

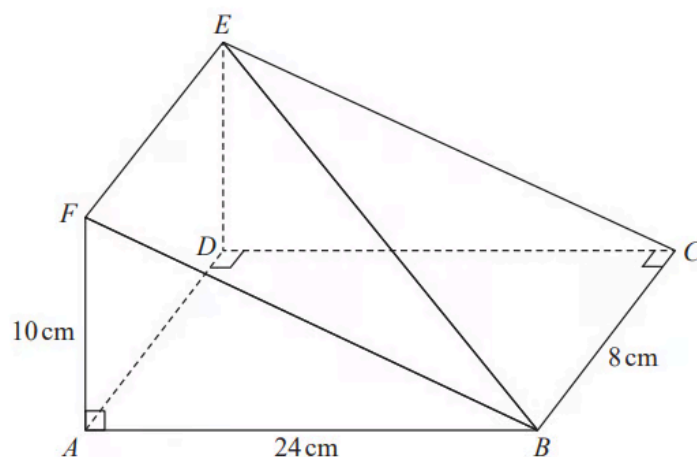


Diagram **NOT**
accurately drawn

$AF = 10 \text{ cm}$, $AB = 24 \text{ cm}$ and $BC = 8 \text{ cm}$.
Angle $FAB = \text{angle } ADC = \text{angle } BCD = 90^\circ$

Work out the size of the angle between the line BE and the plane $ABCD$.
Give your answer correct to 1 decimal place.

(3 marks)

- 5 The diagram shows cuboid $ABCDEFGH$.

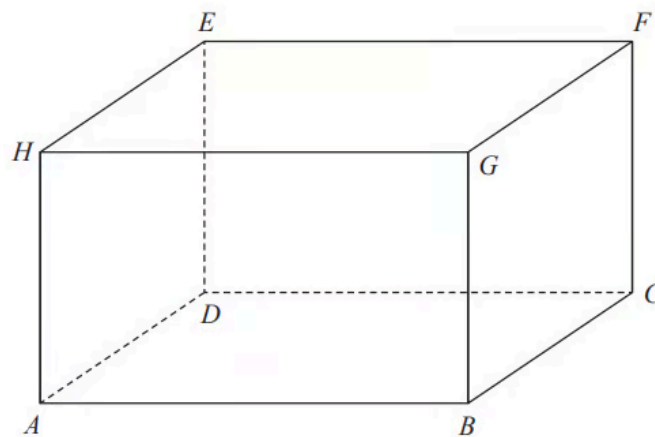


Diagram **NOT**
accurately drawn

For this cuboid

the length of AB : the length of BC : the length of $CF = 4 : 2 : 3$

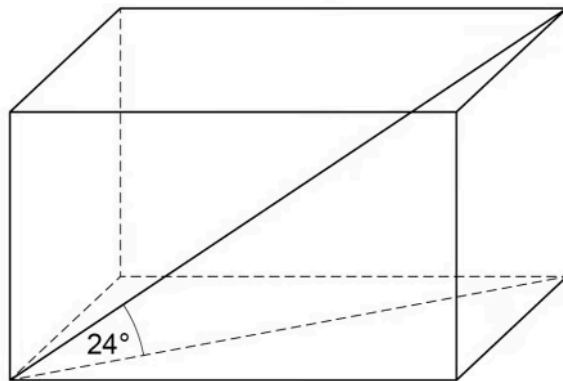
Calculate the size of the angle between AF and the plane $ABCD$. Give your answer correct to one decimal place.

(3 marks)

- 6 The length of a diagonal of a cuboid is 20 cm

The diagonal makes an angle of 24° with the base.

The area of the base is 150 cm^2

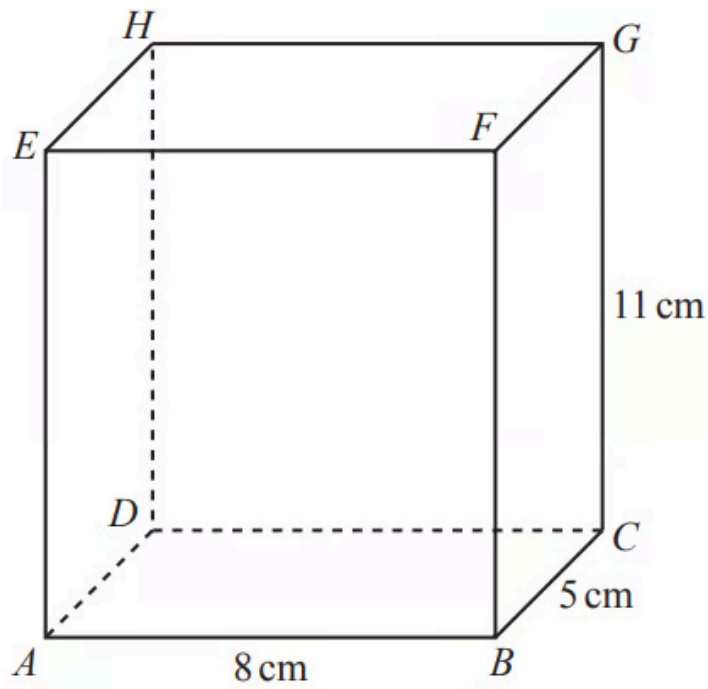


Work out the volume of the cuboid.

..... cm^3

(3 marks)

7



NOT TO
SCALE

$ABCDEFGH$ is a cuboid. $AB = 8$ cm, $BC = 5$ cm and $CG = 11$ cm.

i) Calculate angle CAB .

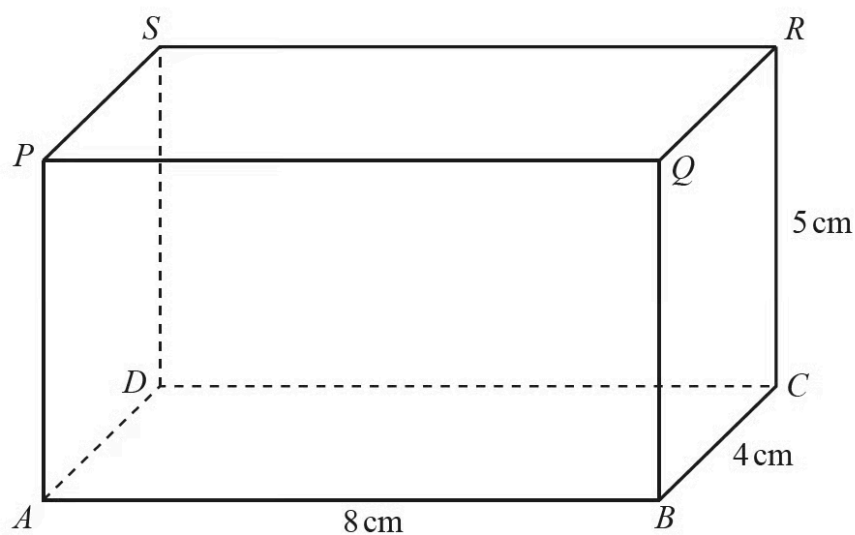
Angle $CAB = \dots\dots\dots$ [2]

ii) Calculate angle GAC .

Angle $GAC = \dots\dots\dots$ [2]

(4 marks)

8



The diagram shows a cuboid.

$AB = 8$ cm, $BC = 4$ cm and $CR = 5$ cm.

Calculate the angle between the diagonal AR and the plane $BCRQ$.

(4 marks)

- 9 $ABCD$ is a horizontal rectangular field.

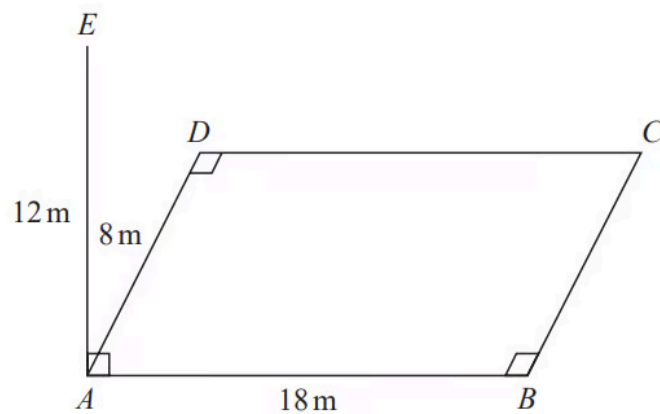


Diagram **NOT** accurately drawn

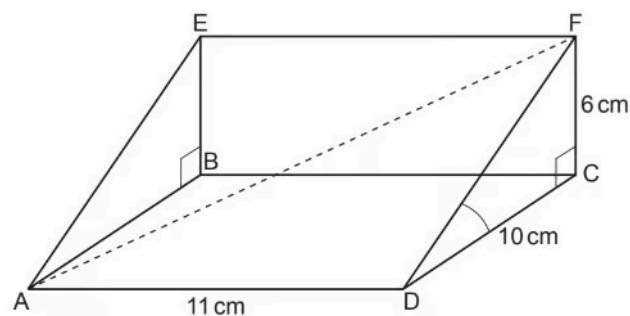
A vertical pole, AE , is placed at the corner A of the field.

$$AE = 12 \text{ m} \quad AB = 18 \text{ m} \quad AD = 8 \text{ m}$$

Calculate the size of the angle between EC and the plane $ABCD$. Give your answer correct to one decimal place.

(3 marks)

- 10 The diagram shows a right-angled triangular prism $ABCDEF$.



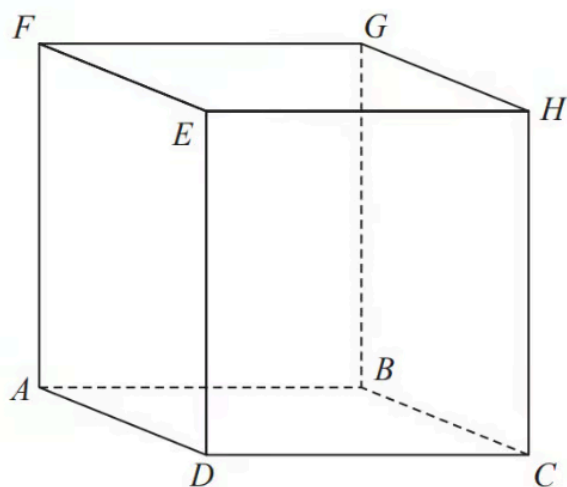
Length $AD = 11 \text{ cm}$, length $CD = 10 \text{ cm}$ and length $CF = 6 \text{ cm}$.

Calculate the exact length of AF .

..... cm

(4 marks)

11 $ABCDEFGH$ is a cuboid.



$$AB = 7.3 \text{ cm}$$

$$CH = 8.1 \text{ cm}$$

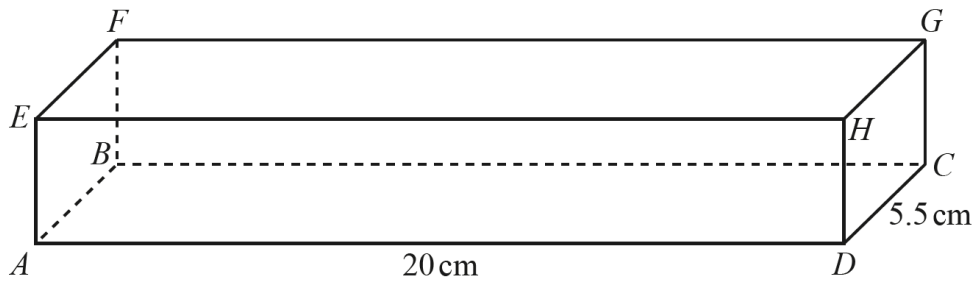
$$\text{Angle } BCA = 48^\circ$$

Find the size of the angle between AH and the plane $ABCD$. Give your answer correct to 1 decimal place.

(4 marks)

Hard Questions

1



NOT TO
SCALE

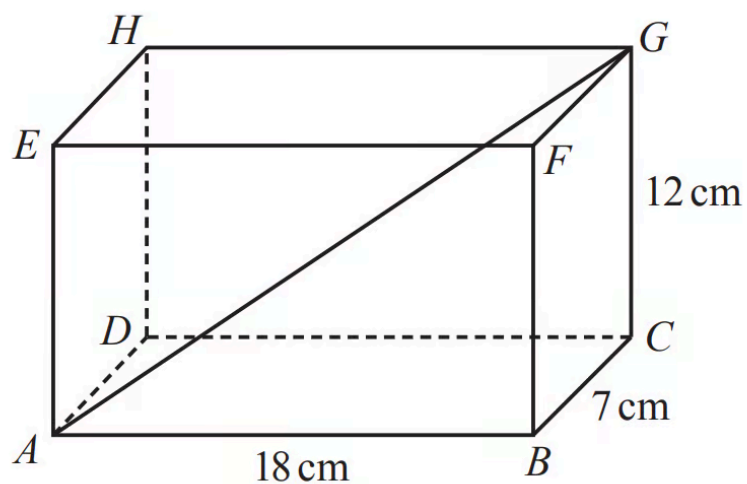
The diagram shows cuboid $ABCDEFGH$ of length 20 cm and width 5.5 cm.

The volume of the cuboid is 495 cm^3 .

Find the angle between the line AG and the base of the cuboid $ABCD$.

(5 marks)

2



NOT TO
SCALE

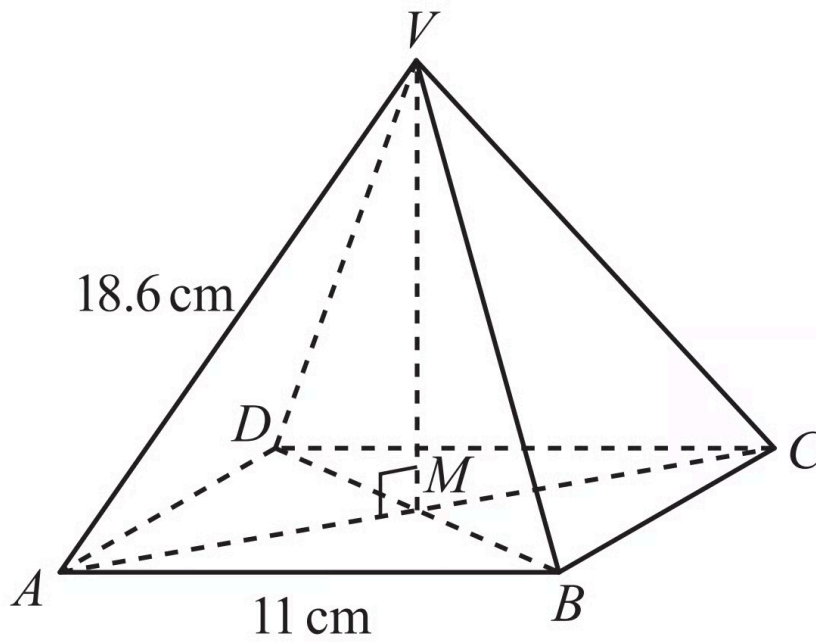
$ABCDEFGH$ is a cuboid.

$AB = 18\text{ cm}$, $BC = 7\text{ cm}$ and $CG = 12\text{ cm}$.

Calculate the angle that the diagonal AG makes with the base $ABCD$.

(4 marks)

3



NOT TO
SCALE

The diagram shows a pyramid with a square base $ABCD$.

The diagonals AC and BD intersect at M .

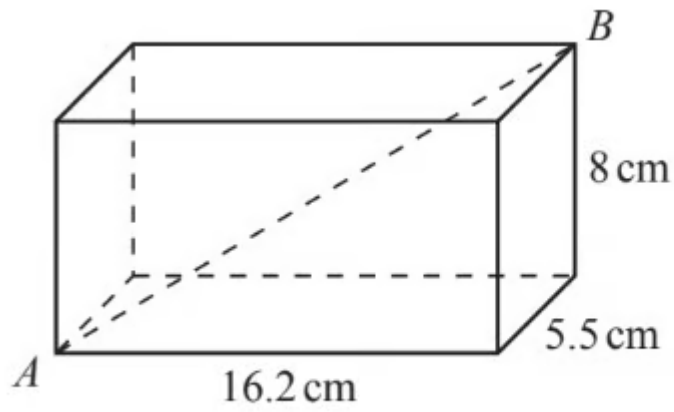
The vertex V is vertically above M .

$AB = 11$ cm and $AV = 18.6$ cm.

Calculate the angle that AV makes with the base.

(4 marks)

4

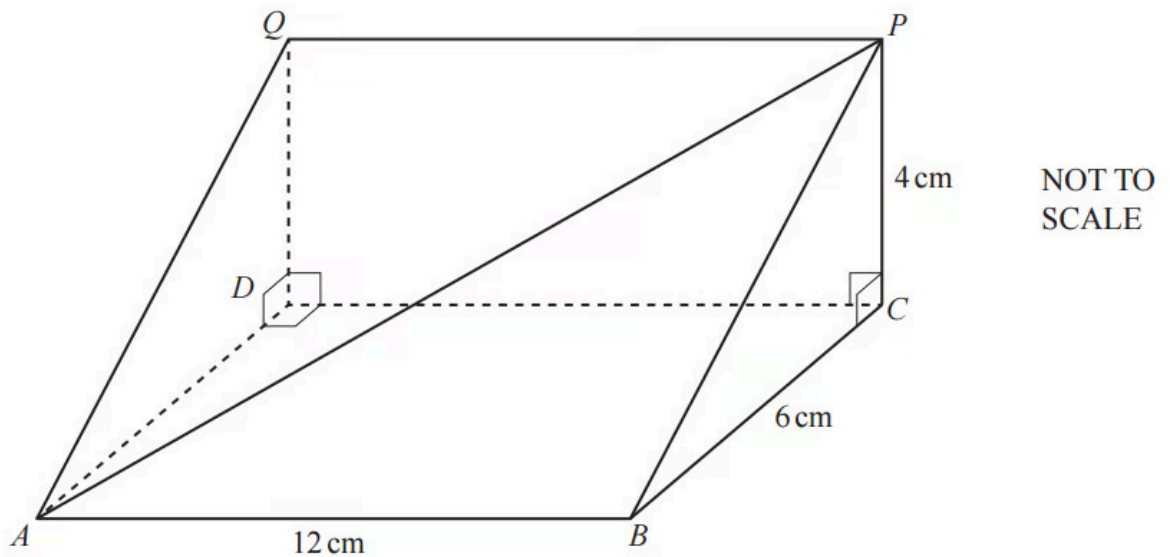


The diagram shows a cuboid with dimensions 5.5 cm, 8 cm and 16.2 cm.

Calculate the angle between the line AB and the horizontal base of the cuboid.

(4 marks)

5



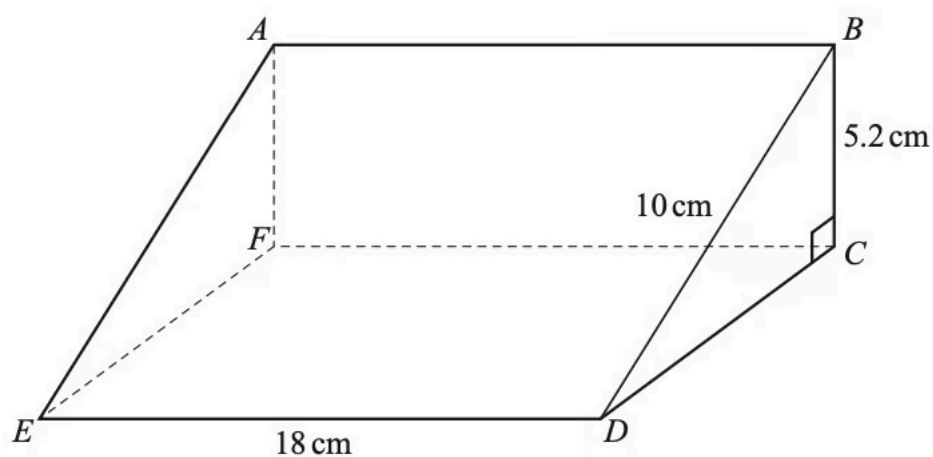
The diagram shows a triangular prism.

$AB = 12$ cm, $BC = 6$ cm, $PC = 4$ cm, angle $BCP = 90^\circ$ and angle $QDC = 90^\circ$.

Calculate the angle between AP and the rectangular base $ABCD$.

(4 marks)

6 (a)



NOT TO
SCALE

The diagram shows a prism $ABCDEF$.

The cross-section is a right-angled triangle BCD .

$BD = 10$ cm, $BC = 5.2$ cm and $ED = 18$ cm.

Calculate angle BEC .

Angle $BEC = \dots\dots\dots$

(4 marks)

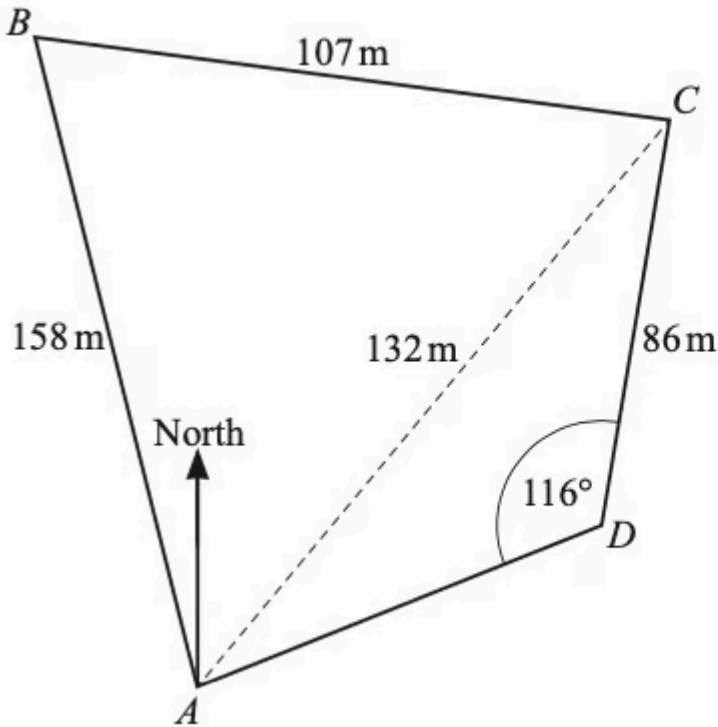
(b) The point G lies on the line ED and $GD = 7$ cm.

Work out angle BGE .

Angle $BGE = \dots\dots\dots$

(3 marks)

7



NOT TO
SCALE

The diagram shows a field, $ABCD$, on horizontal ground.

There is a vertical post at C .

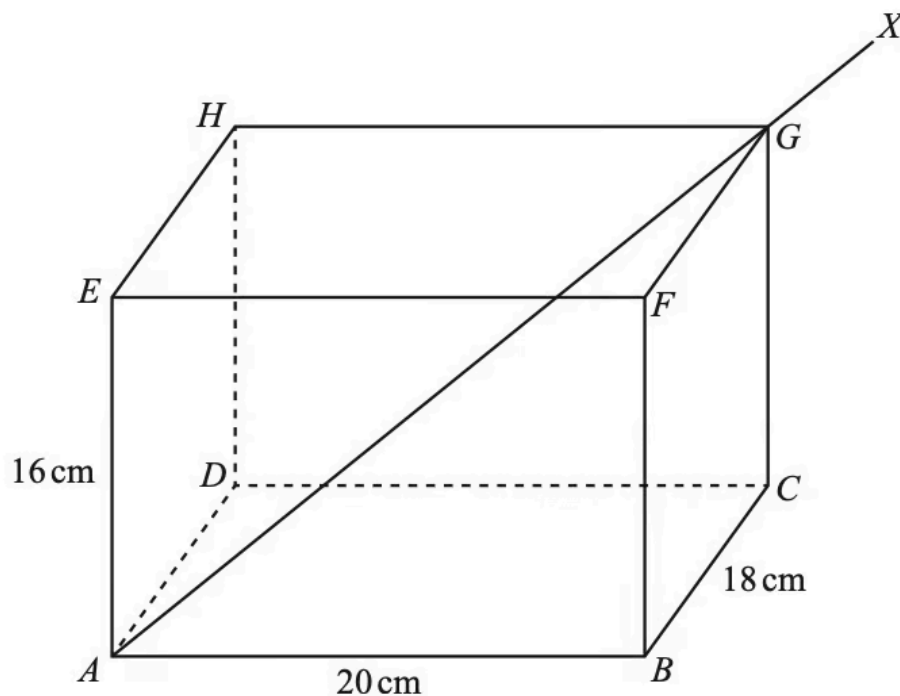
From B , the angle of elevation of the top of the post is 19° .

Find the height of the post.

$\dots\dots\dots$ m

(2 marks)

8 (a)



NOT TO
SCALE

The diagram shows an open box $ABCDEFGH$ in the shape of a cuboid.

$AB = 20$ cm, $BC = 18$ cm and $AE = 16$ cm.

A thin rod AGX rests partly in the box as shown.

The rod is 40 cm long.

Calculate GX , the length of the rod which is outside the box.

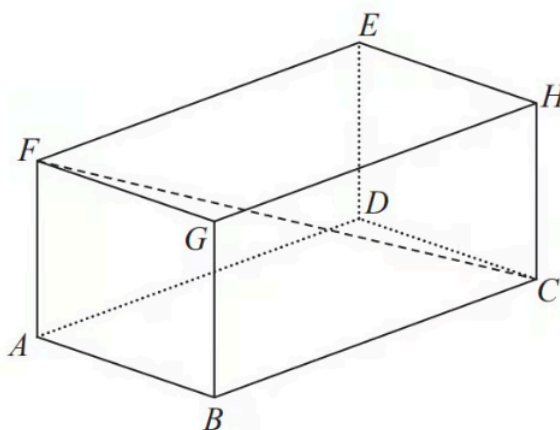
$GX =$ cm

(4 marks)

(b) Calculate the angle the rod makes with the base of the box.

(3 marks)

9 The diagram shows a cuboid $ABCDEFGH$.



$AB = 7$ cm, $AF = 5$ cm and $FC = 15$ cm.

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

(4 marks)

10 (a) The diagram shows a pyramid.

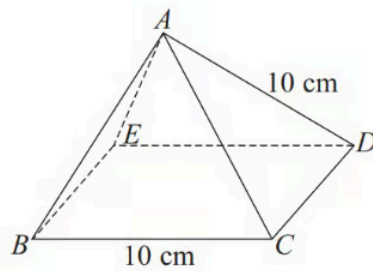


Diagram **NOT**
accurately drawn

$BCDE$ is a square with sides of length 10 cm.

The other faces of the pyramid are equilateral triangles with sides of length 10 cm.

Calculate the volume of the pyramid.

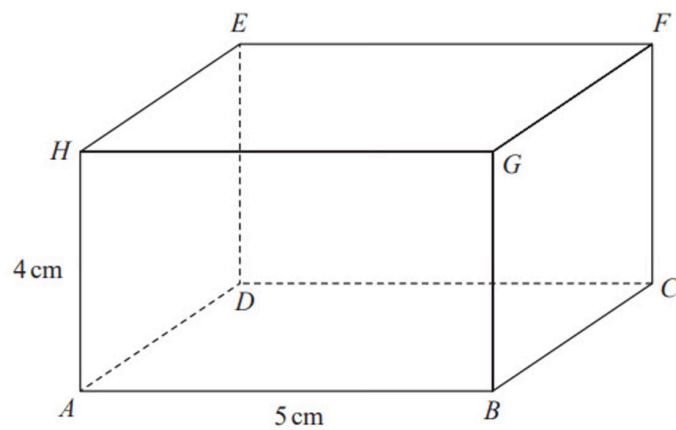
Give your answer in the form $\frac{a}{b}\sqrt{2} \text{ cm}^3$ where a and b are integers to be found.

(4 marks)

(b) Find the size of angle DAB .

(2 marks)

11 The diagram shows a cuboid.



$AB = 5 \text{ cm}$ and $AH = 4 \text{ cm}$.

The size of the angle between CH and the plane $ABCD$ is 35°

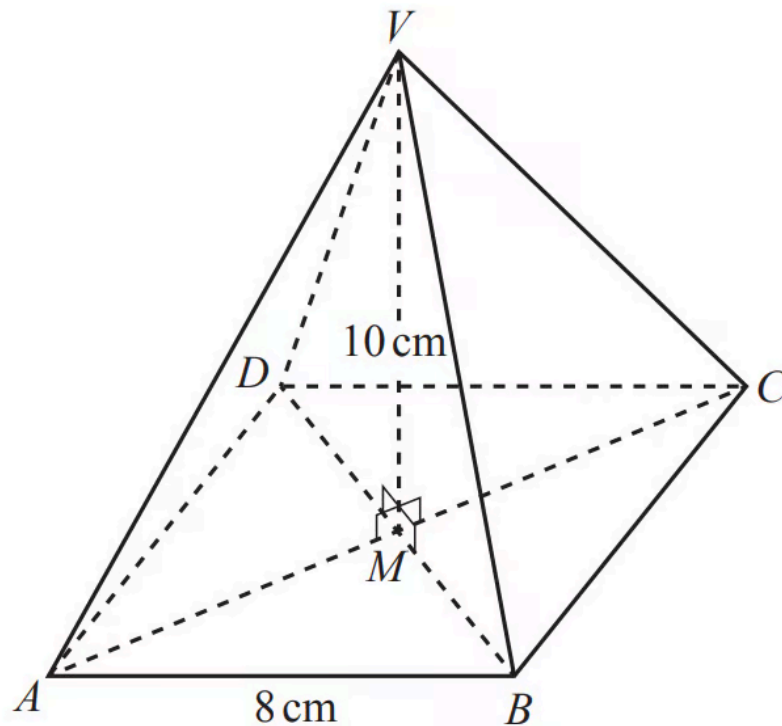
Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

(5 marks)

Very Hard Questions

1



NOT TO
SCALE

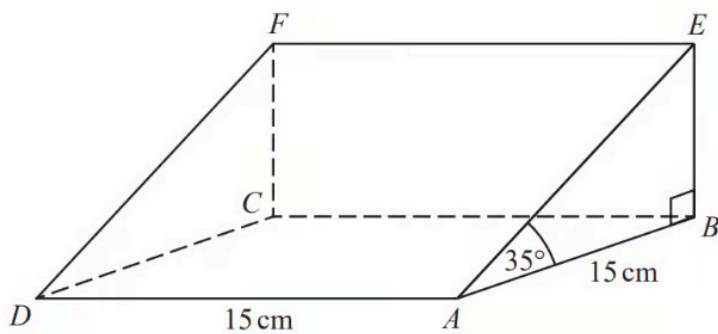
The diagram shows a pyramid with a square base $ABCD$ of side length 8 cm.

The diagonals of the square, AC and BD , intersect at M . V is vertically above M and $VM = 10$ cm.

Calculate the angle between VA and the base.

(4 marks)

2 The diagram shows a triangular prism.



The base, $ABCD$, of the prism is a square of side length 15 cm. Angle ABE and angle CBE are right angles.

Angle $EAB = 35^\circ$

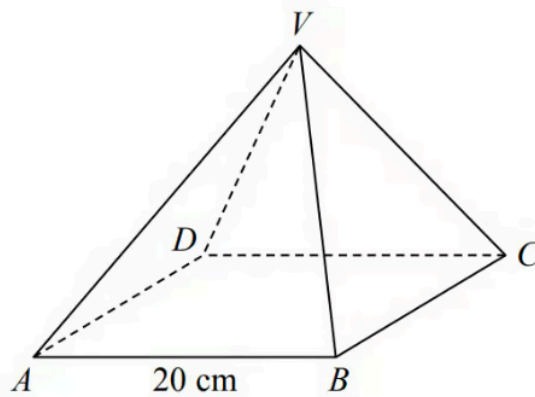
M is the point on DA such that

$$DM:MA = 2:3$$

Calculate the size of the angle between EM and the base of the prism. Give your answer correct to 1 decimal place.

(4 marks)

3 $VABCD$ is a solid pyramid.



$ABCD$ is a square of side 20 cm.

The angle between any sloping edge and the plane $ABCD$ is 55°

Calculate the surface area of the pyramid.

Give your answer correct to 2 significant figures.

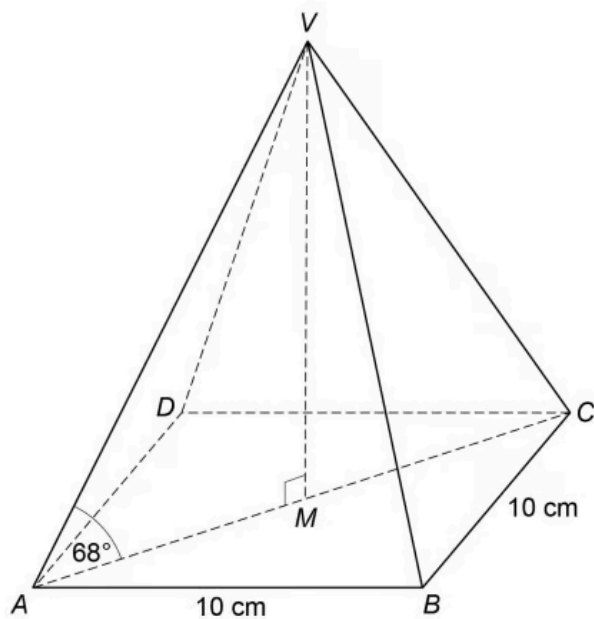
(5 marks)

4 $VABCD$ is a square-based pyramid.

The horizontal base $ABCD$ has side length 10 cm and centre M .

Angle VMA is 90°

Angle VAM is 68°



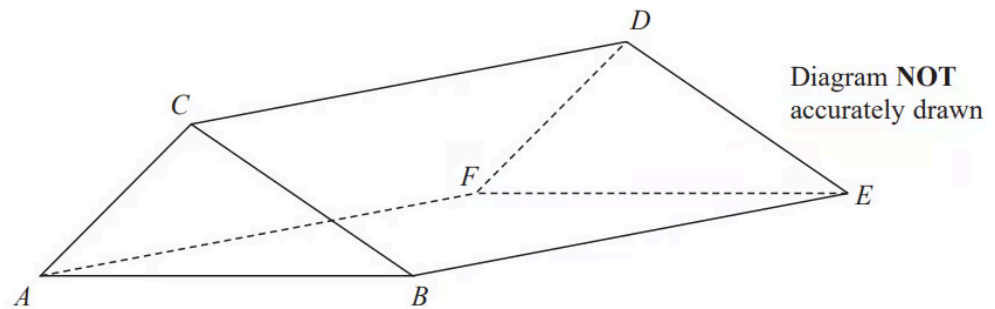
$$\text{Volume of pyramid} = \frac{1}{3} \times \text{area of base} \times \text{perpendicular height}$$

Work out the volume of the pyramid.

..... cm^3

(6 marks)

- 5 The diagram shows the prism $ABCDEF$ with cross section triangle ABC .



Angle $BEC = 40^\circ$ and angle ACB is obtuse.

$AC = 6$ cm and $CE = 13$ cm

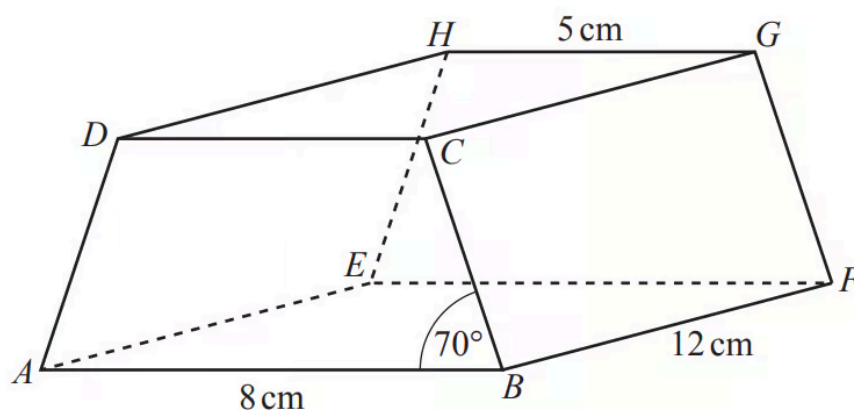
The area of triangle ABC is 22 cm^2

Calculate the length of AB .

Give your answer correct to one decimal place.

.....cm

(6 marks)



NOT TO
SCALE

The diagram shows a prism with a rectangular base, $ABFE$. The cross-section, $ABCD$, is a trapezium with $AD = BC$. $AB = 8$ cm, $GH = 5$ cm, $BF = 12$ cm and angle $ABC = 70^\circ$.

The perpendicular from G onto EF meets EF at X .

i) Show that $EX = 6.5$ cm.

[1]

ii) Calculate AX .

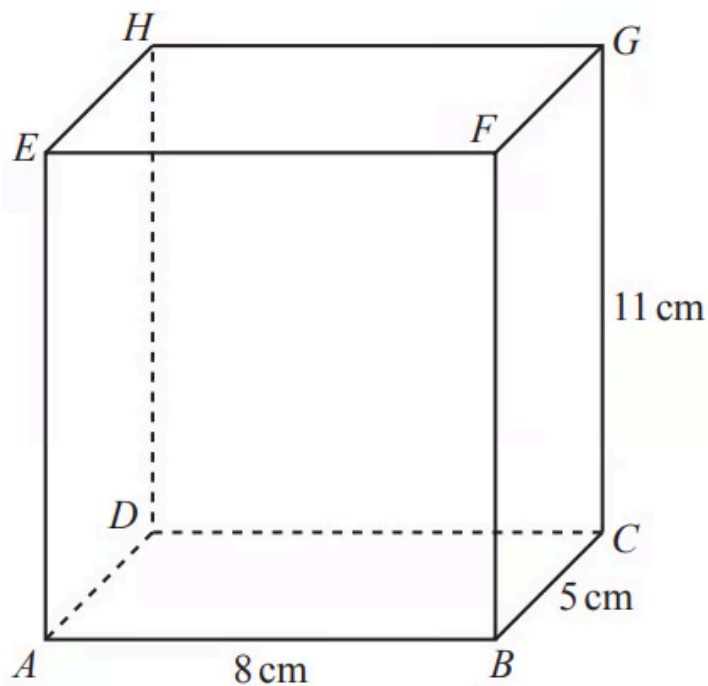
$AX = \dots\dots\dots$ cm [2]

iii) Calculate the angle between the diagonal AG and the base $ABFE$.

[2]

(5 marks)

7



NOT TO
SCALE

$ABCDEFGH$ is a cuboid.

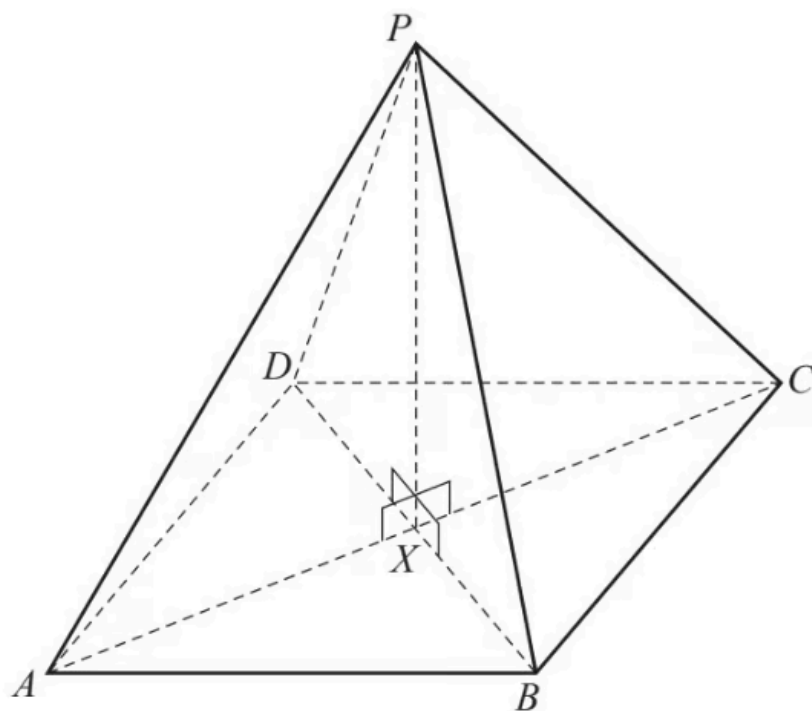
$AB = 8$ cm, $BC = 5$ cm and $CG = 11$ cm.

Ivana has a pencil of length 13cm.

Does this pencil fit completely inside the cuboid?

Show how you decide.

(4 marks)



NOT TO
SCALE

The diagram shows a pyramid with a square base $ABCD$.

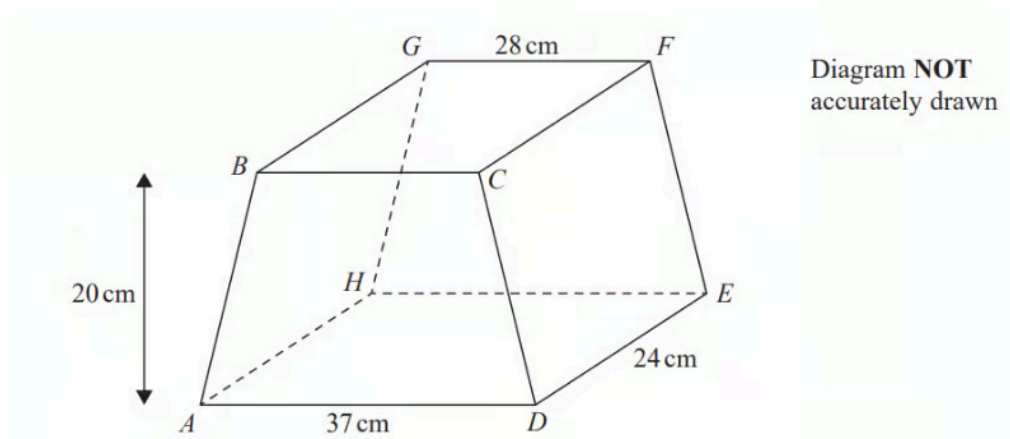
$DB = 8$ cm.

P is vertically above the centre, X , of the base and $PX = 5$ cm.

Calculate the angle between PB and the base $ABCD$.

(3 marks)

9 The diagram shows a solid prism $ABCDEFGH$.



The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

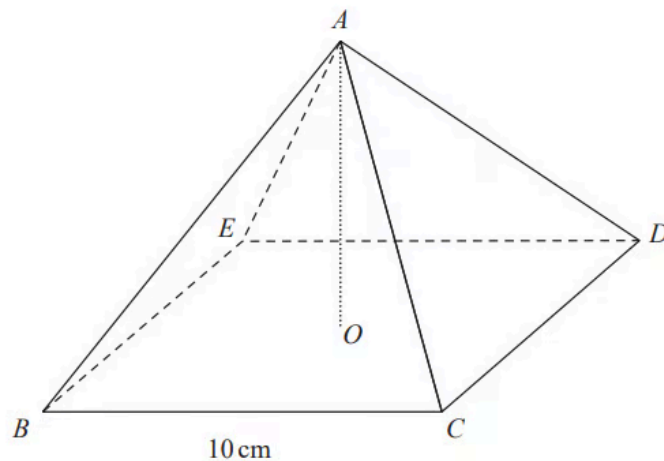
$$AD = 37 \text{ cm} \quad GF = 28 \text{ cm} \quad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

Calculate the size of the angle between AF and the plane $ADEH$. Give your answer correct to one decimal place.

(3 marks)

- 10 The diagram shows a solid pyramid $ABCDE$ with a horizontal base.



The base, $BCDE$, of the pyramid is a square of side 10 cm.

The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$

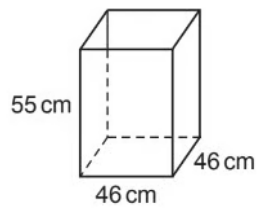
The **total** surface area of the pyramid is 360 cm^2

Work out the size of the angle between AC and the base $BCDE$. Give your answer correct to 3 significant figures.

(6 marks)

- 11 (a) Alvin has a crate in the shape of a cuboid.

The crate is open at the top. The internal dimensions of the crate are 46cm long by 46cm wide by 55cm high.



Alvin has a stick of length 95cm.

Alvin places the stick in the crate so that the shortest possible length extends out above the top of the crate.

Calculate the length of the stick that extends out of the crate.

..... cm

(4 marks)

(b) Calculate the angle the stick makes with the base of the crate.

(3 marks)